



SCEGGS Darlinghurst

2007

**Preliminary Course
Semester 2 Examination**

Mathematics Extension 1

Outcomes Assessed: P1–P8, PE1, PE2, PE3 and PE6
Task Weighting: 40%

General Instructions

- Reading time – 5 minutes
- Working time – 1½ hours
- This paper has **four** questions
- Write using blue or black pen
- Answer all questions on the pad paper provided
- Write your Student Number at the top of each page
- Attempt **all** questions and show all necessary working
- **Start each question on a new page**
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used
- **Do not** attach all questions together in one bundle.

Total marks – 60

- Attempt Questions 1 – 4

Question	Comm	Reason	Calc	TOTAL
1	/3	/2	/3	/15
2	/4	/3	/2	/14
3	/3	/7		/15
4		/10		/16
TOTAL	/10	/22	/5	/60



UNIVERSITY OF BIRMINGHAM

2007

Preliminary Course
Semester 2 Examination

Mathematics Extension 1

Outcomes Assessed: P1-P8, PEB1, PEB2, PEB3 and PEB4
Task Weighting: 40%

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General Instructions

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Total marks – 60

Attempt Questions 1 – 4

Question	Content	Reasoning	Calc	TOTAL
1	A3	10	2	12
2	A4	10	2	12
3	A5	10	2	12
4		10		10
TOTAL	A6	22	2	24

Question 1 (15 marks)

- (a) Solve the inequality: 2

$$x(x - 2) > 3$$

- (b) (i) Fully factorise $f(x) = x^4 + 3x^3 + 2x^2$ 1

- (ii) Sketch the polynomial $y = f(x)$ showing clearly any x and y intercepts. 2
(You do not have to find the co-ordinates of the turning points.)

- (iii) On a new set of axes, sketch the curve $y = f(x - 2)$. 1

- (c) A polynomial is given by $P(x) = x^3 + Ax^2 + Bx - 12$. 3

When $P(x)$ is divided by $(x + 5)$ the remainder is -2 and $(x - 3)$ is a factor of $P(x)$.

Find the values of A and B .

- (d) (i) Write the expansion for $\sin(\theta + \alpha)$. 1

- (ii) If $\tan \theta = \frac{1}{2}$ and $0^\circ < \theta < 90^\circ$, find the exact value of $\sin(\theta + 45^\circ)$ 2

Give your answer in simplest surd form.

- (e) (i) Differentiate $y = \frac{1}{x + a}$. 1

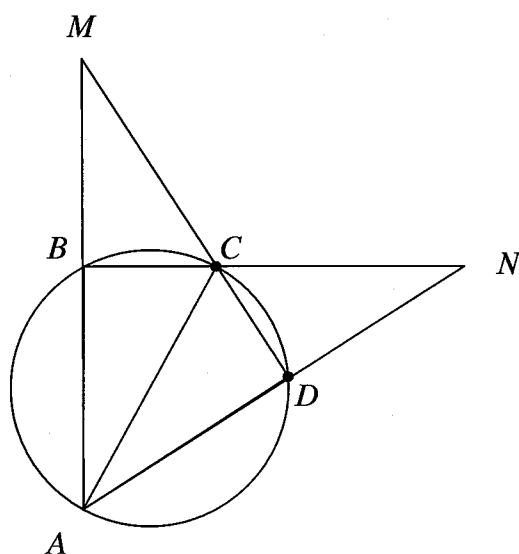
- (ii) Find the value(s) of a if $y = \frac{1}{x + a}$ has a gradient of -1 when $x = 3$. 2

Question 2 (14 marks)

- (a) (i) Find the points of intersection of the hyperbola $y = \frac{3}{x}$ and the line $y = 2x - 5$. 2
- (ii) Sketch the graphs of $y = \frac{3}{x}$ and $y = 2x - 5$ on the same set of axes, showing all important features and points of intersection. 2
- (iii) Let the point where the curves intersect in Quadrant 1 be the point P . Find the gradient of the line and the hyperbola at point P . 2
- (iv) Hence, find the acute angle between the line and the hyperbola at the point P . (Answer correct to the nearest minute.) 2
- (v) Using your graph, or otherwise, find the values of x for which 2

$$2x - 5 \leq \frac{3}{x}$$

(b)



In the diagram, AM , AN , BN and DM are straight lines and $\angle AMD = \angle ANB$.

- (i) By considering $\triangle AMD$ and $\triangle ANB$ or otherwise, explain why $\angle ABC = \angle ADC$. 2
- (ii) Hence prove that AC is a diameter. 2

Question 3 (15marks)

- (a) (i) Express $\cos \theta + \sqrt{3} \sin \theta$ in the form $R \cos(\theta - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$. 2

- (ii) Hence, find the general solutions of $\cos \theta + \sqrt{3} \sin \theta = 1$. 2

- (b) Consider the curve $y = \frac{x^2 + 6}{x + 2}$.

- (i) Use long division to write $\frac{x^2 + 6}{x + 2}$ in the form $Q(x) + \frac{R(x)}{x + 2}$, 2
where $Q(x)$ is the quotient and $R(x)$ is the remainder.

- (ii) Explain why $y = x - 2$ is an oblique asymptote of the graph $y = \frac{x^2 + 6}{x + 2}$. 1

- (iii) State any vertical asymptotes. 1

- (iv) Sketch the curve $y = \frac{x^2 + 6}{x + 2}$ showing any important features. 2
(You do not have to find the co-ordinates of any turning points.)

Question 3 continues on the next page

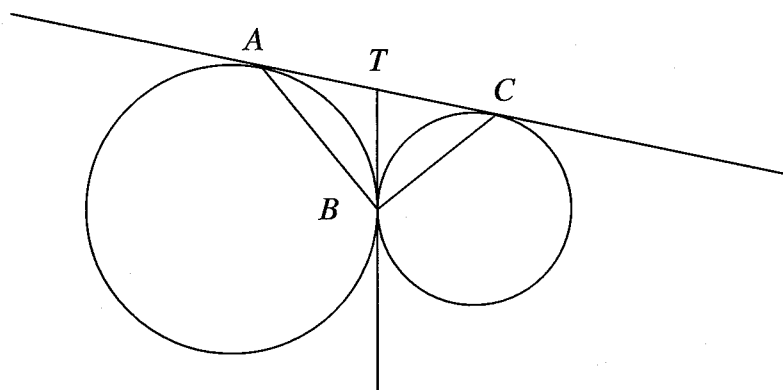
Question 3 (continued)

- (c) TA and TB are two tangents drawn to a circle from an external point T .
 A and B are the points of contact of the tangents with the circle.

Draw a clear diagram on your answer page.

- (i) Prove that $TA = TB$. 3

- (ii) AC is an external tangent to two circles that touch at B . The internal tangent at B meets AC at T . 2



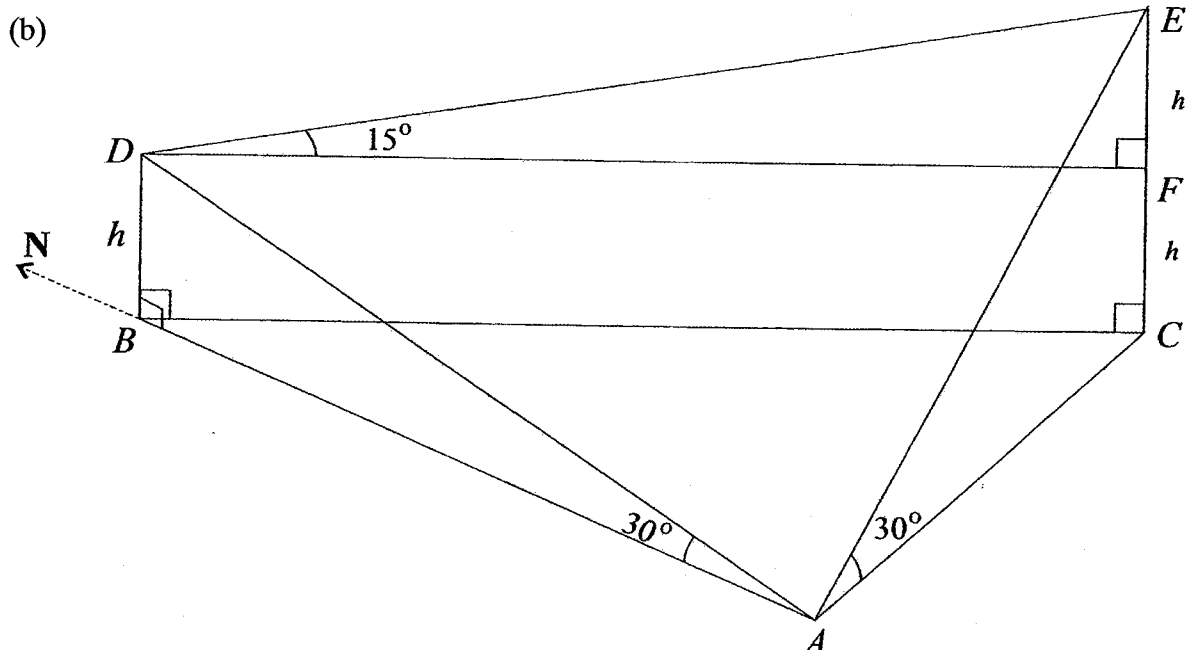
Use part (i) to show T is the centre of the circle passing through A , B and C and hence find the size of $\angle ABC$.

Question 4 (16 marks)

- (a) By making the substitution $t = \tan \frac{\theta}{2}$, prove that

2

$$\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$$



The diagram above shows two vertical towers BD and CE of heights h and $2h$ respectively on a horizontal plane ABC .

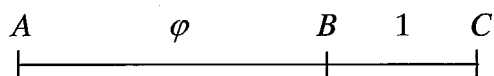
Point A is due south of point B and the angles of elevation of the tops of the towers from A are both 30° . The angle of elevation from D to E is 15° .

- (i) Show that $AB = \sqrt{3}h$. 1
- (ii) Write down expressions for the lengths of AC and BC . 2
- (iii) Hence, find the bearing of the taller tower CE from point A , correct to the nearest degree. 2

Question 4 continues on the next page

Question 4 (continued)

(c)



The diagram shows a straight line segment AC divided by B in the golden ratio $\varphi : 1$.

A divides CB externally in the same ratio that B divides AC internally.

- (i) Show that the golden number φ , satisfies the equation 1

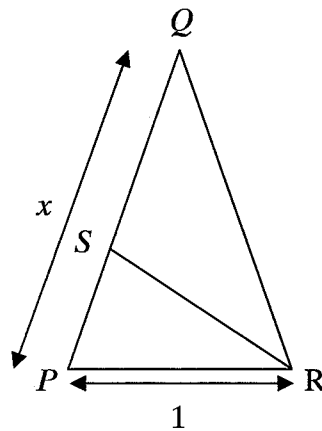
$$\varphi^2 - \varphi - 1 = 0$$

- (ii) Hence find the exact value of φ . 1

Question 4 continues on the next page

Question 4 (continued)

(d)



PQR is an isosceles triangle with $\angle QPR = \angle QRP = 72^\circ$.

$PQ = x$ and $PR = 1$.

The angle bisector of $\angle QRP$ meets QP at S .

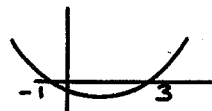
- | | | |
|-------|--|---|
| (i) | Show that $PR = SR = SQ = 1$. | 2 |
| (ii) | Show that $\triangle PQR \parallel \triangle PRS$. | 1 |
| (iii) | Hence show that x is the golden number φ (from part (c)). | 2 |
| (iv) | By using the sine rule in $\triangle PQR$ show that $\cos 36^\circ = \frac{\sqrt{5} + 1}{4}$. | 2 |

End of paper

2007 MATHEMATICS EXT. 1 SEMESTER 2 EXAM (SOLUTIONS)

Question 1

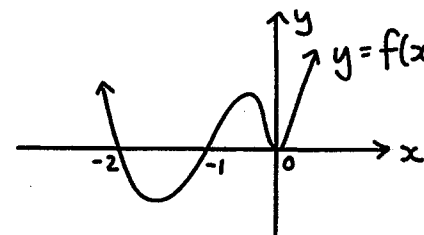
(a) $x(x-2) > 3$
 $x^2 - 2x - 3 > 0$
 $(x-3)(x+1) > 0$ ✓



$x > 3, x < -1$ ✓

(b) i) $f(x) = x^4 + 3x^3 + 2x^2$
 $= x^2(x^2 + 3x + 2)$
 $= x^2(x+1)(x+2)$ ✓

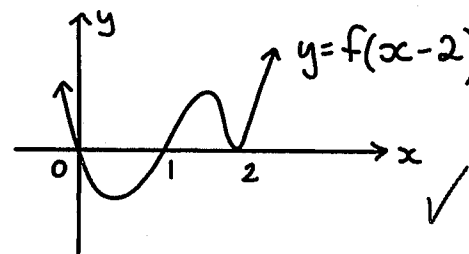
ii)



Comm 3

well done ü

iii)



Shift your part (i)
2 units to the right.

Total /15 Comm /3
Reas /2
Calc /3

Don't forget to expand
first and then form the
quadratic inequality.
You can never do
 $x > 3, x-2 > 3$
Some people got tricked
by this part.

Look for easy steps such
as common factors.
You do not always
need to do long division!

$$(c) P(x) = x^3 + Ax^2 + Bx - 12$$

$$P(-5) = -2 \Rightarrow$$

$$-125 + 25A - 5B - 12 = -2$$

$$25A - 5B = 135$$

$$5A - B = 27 \quad (1) \quad \checkmark$$

$$P(3) = 0 \Rightarrow$$

$$27 + 9A + 3B - 12 = 0$$

$$9A + 3B = -15$$

$$3A + B = -5 \quad (2) \quad \checkmark$$

Solve Simultaneously

$$(1) + (2) \Rightarrow 8A = 22$$

$$A = 2\frac{3}{4}$$

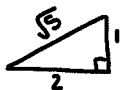
$$\Rightarrow B = -13\frac{1}{4} \quad \checkmark$$

$$(d) i) \sin(\theta + \alpha)$$

$$= \sin\theta \cos\alpha + \cos\theta \sin\alpha \quad \checkmark$$

$$ii) \tan\theta = \frac{1}{2} \Rightarrow$$

$$0^\circ < \theta < 90^\circ$$



$$\sin(\theta + 45^\circ)$$

$$= \sin\theta \cos 45^\circ + \cos\theta \sin 45^\circ$$

$$= \frac{1}{\sqrt{5}} \times \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{2}} \quad \checkmark$$

$$= \frac{3}{\sqrt{10}}$$

$$= \frac{3\sqrt{10}}{10} \quad \checkmark$$

An easy question!

Learn your theorems!

By the remainder theorem.

$$P(-5) = -2$$

By the factor theorem,

$$P(3) = 0$$

Note: In algebra A is different to a.
B is different to b.
Do not mix capitals and small letters.

Well done. Everyone knew this rule ü

Reas. 2

Many students didn't see the link in this question. Why are you given $\tan\theta = \frac{1}{2}$?

This part stumped a lot of students. It's not really too hard.

→ You must show the substitution for $\sin\theta$ and $\cos\theta$ to get this mark.

Final answer must have rationalised denominator. Read the fine details of the question.

$$(e) i) y = \frac{1}{(x+a)}$$

$$= (x+a)^{-1}$$

Use the Chain rule to differentiate

$$\frac{dy}{dx} = -1(x+a)^{-2} \times 1$$

$$= \frac{-1}{(x+a)^2} \quad \checkmark$$

Calc. 3

Note that a is just a constant. (It works like a number.)
Using the quotient rule is open to more errors.

$$ii) \frac{dy}{dx} = -1 \text{ when } x = 3$$

$$\Rightarrow -1 = \frac{-1}{(3+a)^2}$$

$$(3+a)^2 = 1$$

$$3+a = \pm 1 \quad \checkmark$$

$$\begin{array}{l|l} 3+a=1 & 3+a=-1 \\ a=-2 & a=-4 \end{array} \quad \checkmark$$

or you could expand this quadratic and then solve it
 $9 + 6a + a^2 = 1$
 $a^2 + 6a + 8 = 0$
etc.

Question 2

(a) i) For pts of int. solve simult.

$$\frac{3}{x} = 2x - 5$$

$$3 = 2x^2 - 5x$$

$$2x^2 - 5x - 3 = 0$$

$$2x^2 - 6x + x - 3 = 0$$

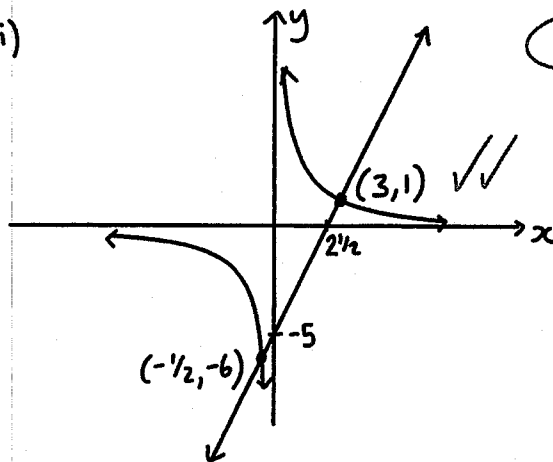
$$(2x + 1)(x - 3) = 0$$

$$x = -\frac{1}{2} \quad x = 3 \quad \checkmark$$

$$y = -6 \quad y = 1 \quad \checkmark$$

$\therefore$ Pts of int: $(-\frac{1}{2}, -6), (3, 1)$

ii)



Comm 2.

* All points of intersection and x/y intercepts should be labelled - an indistinct straight line for $y = 2x - 5$ was not good enough.

* Hyperbolas weren't particularly well drawn - it is important they approach the asymptotes

iii) m of line = 2 $\checkmark$

m of hyperbola: $y = \frac{3}{x}$
 $y' = -\frac{3}{x^2}$

$$m_T = -\frac{3}{3^2} = -\frac{1}{3} \quad \checkmark$$

Calc 2.

Differentiation of $\frac{3}{x}$ & subsequent substitution of $x = 3$ was extremely poorly done!

Total /14
Comm /4
Reas /2
Calc /2

Nicely done

$$\begin{aligned} \text{iv) } \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{2 - (-\frac{1}{3})}{1 + (-\frac{2}{3})} \right| \quad \checkmark \\ &= 7 \end{aligned}$$

$$\theta = \tan^{-1} 7$$

$$= 81^\circ 52' \text{ (to nearest minute)} \quad \checkmark$$

$$\text{v) } x \leq -\frac{1}{2} \quad \checkmark, \quad 0 < x \leq 3 \quad \checkmark$$

Most new this formula & how to apply it

It is easiest to read the solution off the graph. Most chose to solve 'otherwise' and did so poorly - must remember $x \neq 0$ & also to multiply by the denominator squared

$$\begin{aligned} \text{(b) i) } \angle AMD &= \angle ANB \text{ (given)} \\ \angle MCB &= \angle NCD \\ \text{(vertically opposite } \angle s &=) \\ \therefore \angle AMD + \angle MCB &= \angle ANB + \angle NCD \\ \Rightarrow \angle ABC &= \angle ADC \\ \text{(exterior } \angle \Delta &= \text{sum of} \\ &\text{2 opposite interior } \angle s) \end{aligned}$$

$$\begin{aligned} \text{ii) } \angle ABC &= \angle ADC \text{ (above)} \\ \angle ABC + \angle ADC &= 180^\circ \\ \text{(opposite } \angle s \text{ of a cyclic} &\text{quad. are supplementary)} \\ \therefore \angle ABC &= \angle ADC = 90^\circ \\ \therefore AC &\text{ is a diameter} \\ \text{(} \angle s \text{ in a semicircle} &= 90^\circ) \quad \checkmark \end{aligned}$$

Comm 2

$\checkmark$ Many proved similar Δ s using just 2 angles then quoted the 3rd angles as equal because of corresponding angles ...
 $\rightarrow$ This shows you

Reas 2 don't really understand why only 2 angles are required to prove similarity! The 3rd pair of angles are = because $\angle \text{sum } \Delta = 180^\circ!$

Part ii was done well by those who attempted it

Question 3

(a) i) $\cos\theta + \sqrt{3}\sin\theta \equiv R\cos(\theta - \alpha)$
 $\equiv R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$
 $R\cos\alpha = 1$ ①
 $R\sin\alpha = \sqrt{3}$ ②

Find α
 ② $\frac{R\sin\alpha}{R\cos\alpha} = \frac{\sqrt{3}}{1}$
 $\tan\alpha = \sqrt{3}$
 $\alpha = 60^\circ$ ✓

Find R
 $R^2 = 1^2 + \sqrt{3}^2$
 $R = \sqrt{1+3}$
 $= \sqrt{4}$
 $= 2$ ✓

$\therefore \cos\theta + \sqrt{3}\sin\theta = 2\cos(\theta - 60^\circ)$

ii) $\cos\theta + \sqrt{3}\sin\theta = 1$
 $2\cos(\theta - 60^\circ) = 1$
 $\cos(\theta - 60^\circ) = \frac{1}{2}$ (Quad 1 & 4)
 $\theta - 60^\circ = -60^\circ, 60^\circ, 300^\circ$
 $\theta = 0^\circ, 120^\circ, 360^\circ$

$(\theta - 60^\circ) = 60^\circ + 360^\circ n, -60^\circ + 360^\circ n$
 $\theta = 120^\circ + 360^\circ n, 360^\circ n$
 for n an integer

Total /15 Comm /3
Reas /7

This part was very well done. Everyone seems to have mastered this technique.

Note the domain
 if $0^\circ \leq \theta \leq 360^\circ$
 $-60^\circ \leq \theta - 60^\circ \leq 300^\circ$

Reas 2

For the general solution look at the pattern. Add revolutions to these angles that you found here.

* Don't mix radians and degrees.

(b) i)
$$\begin{array}{r} x-2 \\ x+2 \overline{) x^2 + 6} \\ \underline{x^2 + 2x} \\ -2x + 6 \\ \underline{-2x - 4} \\ 10 \end{array}$$

$\therefore \frac{x^2+6}{x+2} = (x-2) + \frac{10}{x+2}$ ✓

ii) $y = (x-2) + \frac{10}{x+2}$

as $x \rightarrow \infty$, $\frac{10}{x+2} \rightarrow 0$ ✓

$\therefore y = x-2$ is an oblique asymptote

Long division was well done except for a few silly addition mistakes.

$$\begin{array}{r} Q(x) \\ A(x) \overline{) P(x)} \\ \hline R(x) \end{array}$$

Note the correct position of the parts.

Comm 1

You must state clearly that $\frac{10}{x+2}$ approaches zero to get this mark. Just making it disappear from one line to the next is not sufficient.

iii) Vertical asymptote: $x = -2$ ✓

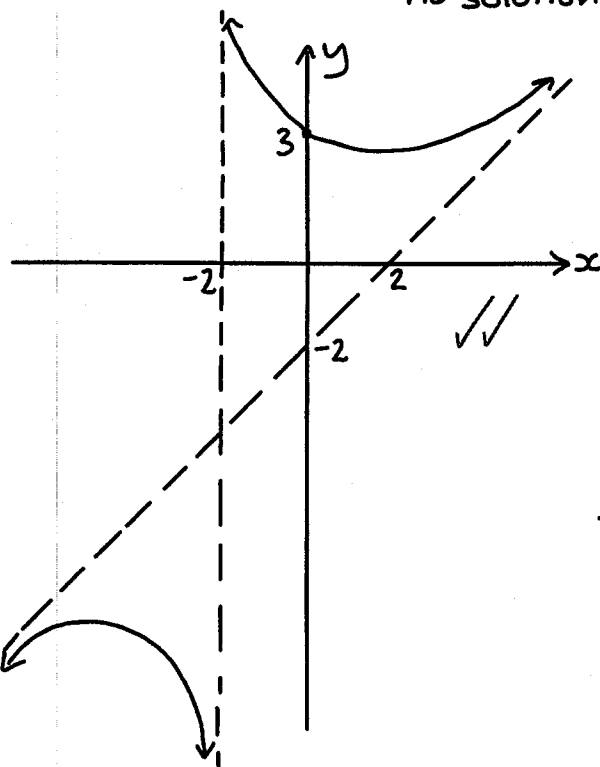
The vertical asymptote is a line. It is incorrect to say that its equation is $x \neq -2$.

iv) y int: $x=0 \rightarrow y=3$

x int: $y=0 \rightarrow \frac{x^2+6}{x+2} = 0$

$x^2+6=0$

no solutions



Comm 2

There were some problems with connecting the ideas in the previous parts.

For $y = \frac{x^2+6}{x+2}$

Think!

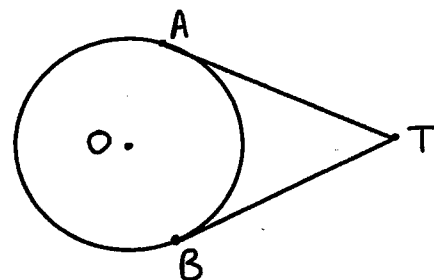
- there are no possible x intercepts.

$\begin{pmatrix} x^2+6=0 \\ y=0 \\ \text{No solution} \end{pmatrix}$

- The y intercept is (0,3).

- Draw a sign diagram or check some points for various x values to check your graph.

(c)



Reas 3

Very poorly done!
You are required to prove why $TA=TB$. part (i) is worth 3 marks so you have to do more than just quote a theorem.

Here's one method.

- i) Let O be the centre of the circle

In $\triangle OAT$ & $\triangle OBT$

$OA = OB$ (equal radii of circle)

$OT = OT$ (common)

$\angle OAT = \angle OBT = 90^\circ$

(radius $\perp$ tangent at the point of contact) ✓

$\therefore \triangle OAT \equiv \triangle OBT$ (RHS) ✓ (proof using congruent $\triangle$ s)

$\therefore TA = TB$ (matching sides in congruent $\triangle$ s =) ✓

Here's another method.

Mark in point C on circumference.

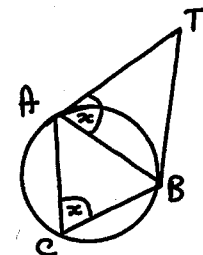
$\angle TAB = \angle ACB$ ($\angle$ between a tangent and a chord = $\angle$ in the alternate segment)

Similarly,

$\angle TBA = \angle ACB$ ($\angle$ between tangent and chord = $\angle$ in the alternate segment)

$\therefore \angle TAB = \angle TBA$

$\therefore TA = TB$ (sides opposite equal $\angle$ s are equal)



ii) $TA = TB = TC$ (using part i)
 $\therefore T$ is the centre of the
 circle passing through A, B, C ✓

Since T lies on chord AC & $TA = TC$,
 AC is the diameter
 $\therefore \angle ABC = 90^\circ$
 ($\angle$ in a semicircle = 90°) ✓

Reas 2

This part was
 done better than
 part (i). Make
 sure to answer
 both parts.

Question 4

$$\begin{aligned}
 \text{(a) LHS} &= \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} \\
 &= \frac{\frac{1+t^2}{1+t^2} + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}{\frac{1+t^2}{1+t^2} + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \quad \checkmark \\
 &= \frac{1+t^2+2t-1+t^2}{1+t^2+2t+1-t^2} \\
 &= \frac{2t^2+2t}{2+2t} \\
 &= \frac{2t(t+1)}{2(1+t)} \quad \checkmark \\
 &= t \\
 &= \tan \theta/2 \\
 &= \text{RHS}
 \end{aligned}$$

Total /16 Reas /10

Most substituted the
 correct t-formulae into LHS
 however algebra was
 extremely inefficient
 and often poorly done.

$$\begin{aligned}
 \text{(b) i) In } \triangle ABD \\
 \tan 30^\circ &= \frac{h}{AB} \\
 \frac{1}{\sqrt{3}} &= \frac{h}{AB} \\
 AB &= \sqrt{3}h \quad \checkmark
 \end{aligned}$$

An easy mark scored
 by most.

ii) Similarly $AC = \sqrt{3} \times 2h$
 $AC = 2\sqrt{3}h$ ✓

In $\triangle DEF$
 $\tan 75^\circ = \frac{DF}{h}$
 $DF = h \tan 75^\circ$
 $\therefore BC = h \tan 75^\circ$ ✓

Generally done well.

Some tried to find BC using Pythagoras in $\triangle ABC$ - this does not work because $\triangle ABC$ is not right angled

iii) $\cos(\angle BAC) = \frac{b^2 + c^2 - a^2}{2bc}$
 $= \frac{(\sqrt{3}h)^2 + (2\sqrt{3}h)^2 - (h \tan 75^\circ)^2}{2(\sqrt{3}h)(2\sqrt{3}h)}$
 $= \frac{3h^2 + 12h^2 - h^2 \tan^2 75^\circ}{12h^2}$
 $= \frac{15 - \tan^2 75^\circ}{12}$
 $= 0.089 \dots$

Reas. 2

Many had the correct idea but a badly executed solution with poor algebra

$\angle BAC = 85^\circ$ (to nearest degree)

$\therefore$ bearing of CE from A is $085^\circ T$

You must ensure you answer the question!

(c) i) $\frac{\theta+1}{\theta} = \frac{\theta}{1}$

Reas 2

$\theta+1 = \theta^2$
 $\theta^2 - \theta - 1 = 0$ ✓

ii) $\theta = \frac{1 \pm \sqrt{5}}{2}$

since θ is a +ve distance
 $\theta = \frac{1 + \sqrt{5}}{2}$ ✓

* A straight forward solution made a mess of by most
 * Note that if the word "externally" is used in the question, a negative is not required in the ratio.

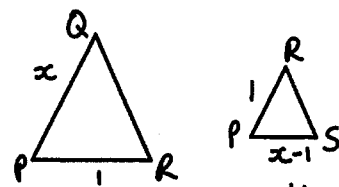
Ext. 1 candidates & indeed 2 unit candidates MUST be able to solve a simple quadratic ... blindfolded!

(d) i) $\angle QPR = 72^\circ$ (given)
 $\angle SRQ = \angle PRS = 36^\circ$ (RS bisects $\angle QRP$)
 $\therefore \angle PSR = 72^\circ$ (L sum $\triangle PSR = 180^\circ$)
 $\therefore SR = PR = 1$ (Ls opposite = sides of $\triangle PSR =$)
 $\angle SQR = 36^\circ$ (L sum $\triangle PQR = 180^\circ$)
 $SQ = SR = 1$ (Ls opposite = sides of $\triangle SQR =$) ✓

Reas 2

Well done, however the proofs could have been more succinct.

ii) In $\triangle PQR$ & $\triangle PRS$
 $\angle QPR = \angle RPS$ (common)
 $\angle QRP = \angle RSP = 72^\circ$ (above)
 $\therefore \triangle PQR \sim \triangle PRS$ (AA similarity test) ✓

iii) 
 $\frac{x}{1} = \frac{1}{x-1}$ (corresponding sides in similar $\triangle$ s are in the same ratio)
 $x^2 - x = 1$
 $x^2 - x - 1 = 0$ ✓
 $\therefore x = \frac{1 + \sqrt{5}}{2}$ which is φ

Reas 2

This could have easily been done with or without ii. Not a single student got a mark in iii! "Hence" after a similar $\triangle$ s proof almost always means use "matching sides in similar $\triangle$ s in the same ratio"

iv) Sin Rule in $\triangle PQR$
 $\frac{x}{\sin 72^\circ} = \frac{1}{\sin 36^\circ}$ ✓
 $x \sin 36^\circ = \sin 72^\circ$
 $x \sin 36^\circ = 2 \sin 36^\circ \cos 36^\circ$
 $\cos 36^\circ = \frac{x}{2}$
 $= \frac{\sqrt{5} + 1}{4}$ ✓

Reas 2

Once again these 2 marks are accessible even if other parts were skipped. Simply follow the instructions.

Could also get this result using cos Rule in $\triangle PQR$